Solutions for Week 9 Maths Exercise Sheet
Vector Spaces: Definition, Examples, Span, Linear (in)dependence

(1) Consider the vector space $\mathbb{Q}^2$ of 2-tuples of rational numbers over the field Q of rational numbers. Show that if $\vec{u} = \begin{pmatrix}1\\0\end{pmatrix}$ and $\vec{v} = \begin{pmatrix}1\\1\end{pmatrix}$ then $\text{Span}(\vec{u}, \vec{v}) = \mathbb{Q}^2$.

Comment: You need to show that $\begin{pmatrix}a\\b\end{pmatrix} \in \text{Span}(\vec{u}, \vec{v})$ for every $a, b \in \mathbb{Q}$.

Solution: We want to show that $\vec{u}$ and $\vec{v}$ span each vector in $\mathbb{Q}^2$. How does a general vector in $\mathbb{Q}^2$ look like? It is of the form $\binom{a}{b}$ for some $a, b \in \mathbb{Q}$. We now want to show that there exist $r, s \in \mathbb{Q}$ such that

$$r\vec{u} \oplus s\vec{v} = r\begin{pmatrix}1\\0\end{pmatrix} \oplus s\begin{pmatrix}1\\1\end{pmatrix} = \begin{pmatrix}a\\b\end{pmatrix}$$

Equating the first co-ordinate we get

$$r + s = a \tag{1}$$

Equating the second co-ordinate we get

$$\mathbf{s = b} \tag{2}$$

Substituting $s = b$ in equation (1) gives $\mathbf{r} = (\mathbf{a} - \mathbf{b})$. We verify that our values of r, s are correct by checking that

$$(a-b)\begin{pmatrix}1\\0\end{pmatrix} \oplus b\begin{pmatrix}1\\1\end{pmatrix} = \begin{pmatrix}a-b\\0\end{pmatrix} \oplus \begin{pmatrix}b\\b\end{pmatrix} = \begin{pmatrix}a\\b\end{pmatrix}$$

---

(2) Consider the vector space $\mathbb{Q}^3$ of 3-tuples of rational numbers over the field Q of rational numbers. Show that $\vec{w} \notin \text{Span}(\vec{u}, \vec{v})$ where $\vec{w} = \begin{pmatrix}9\\27\\35\end{pmatrix}$, $\vec{u} = \begin{pmatrix}1\\3\\4\end{pmatrix}$ and $\vec{v} = \begin{pmatrix}2\\6\\8\end{pmatrix}$.

Solution: We prove this by contradiction.

Suppose there exists $r, s \in \mathbb{Q}$ such that $r\vec{u} \oplus s\vec{v} = \vec{w}$, i.e.,

$$r\begin{pmatrix}1\\3\\4\end{pmatrix} \oplus s\begin{pmatrix}2\\6\\8\end{pmatrix} = \begin{pmatrix}9\\27\\35\end{pmatrix}$$

Equating the first co-ordinate we get

$$r + 2s = 9 \tag{3}$$

Equating the second co-ordinate we get

$$3r + 6s = 27 \tag{4}$$

Equating the third co-ordinate we get

$$4r + 8s = 35 \tag{5}$$

Adding equation (3) and equation (4) gives

$$4r + 8s = 36 \tag{6}$$

which contradicts equation (5). Hence, there are no rational numbers $r, s \in \mathbb{Q}$ such that $r\vec{u} \oplus s\vec{v} = \vec{w}$, i.e., $\vec{w} \notin \text{Span}(\vec{u}, \vec{v})$

---

(3) Let V be a vector space over a field F. Let $\vec{u}, \vec{v}$ be any two vectors in V. Recall that $\text{Span}(\vec{u}, \vec{v}) = \{r\vec{u} \oplus s\vec{v} \mid r, s \in F\}$. Verify that $\text{Span}(\vec{u}, \vec{v})$ is a vector space over F.

Comment: First define for $\text{Span}(\vec{u}, \vec{v})$ the two operations of (i) vector addition, and (ii) multiplication of a scalar with a vector. Each of the 8 conditions in the definition of vector space over a field now needs to be verified. You can use any of the 8 conditions available due to V being a vector space over the field F.
Solution: First, we define two operations of vector addition and multiplication of a scalar with a vector for the vector space $\text{Span}(\vec{u}, \vec{v})$ as follows:

- Definition of vector addition: Any vector in $\text{Span}(\vec{u}, \vec{v})$ is of the form $r\vec{u} \oplus s\vec{v}$ for some $r, s \in F$. Note that we have $+$ which is used to denote addition of two scalars in F, and $\oplus$ which is used to denote addition of two vectors from V. We will again use $\oplus$ to denote addition of vectors from $\text{Span}(\vec{u}, \vec{v})$ since each element/vector of $\text{Span}(\vec{u}, \vec{v})$ is a vector from V. The vector addition for $\text{Span}(\vec{u}, \vec{v})$ is defined as follows:

$$(r_1\vec{u} \oplus s_1\vec{v}) \oplus (r_2\vec{u} \oplus s_2\vec{v}) = r_1\vec{u} \oplus s_1\vec{v} \oplus r_2\vec{u} \oplus s_2\vec{v}$$

- Definition of multiplication of a scalar with a vector: Any vector in $\text{Span}(\vec{u}, \vec{v})$ is of the form $r\vec{u} \oplus s\vec{v}$ for some $r, s \in F$. Multiplication of this vector with a scalar is defined using the operation of multipliying a scalar with a vector from V as follows:

$$t(r\vec{u} \oplus s\vec{v}) = tr\vec{u} \oplus ts\vec{v}$$

We now verify the 8 conditions for $\text{Span}(\vec{u}, \vec{v})$ to be a vector space over the field F. Let $\vec{x} = r_1\vec{u} \oplus s_1\vec{v}$, $\vec{y} = r_2\vec{u} \oplus s_2\vec{v}$ and $\vec{z} = r_3\vec{u} \oplus s_3\vec{v}$ be any three vectors from $\text{Span}(\vec{u}, \vec{v})$ where $r_1, r_2, r_3, s_1, s_2, s_3$ are scalars in F. Let t, q be any scalars in F.

(1) Commutativity of vector addition:
$\vec{x} \oplus \vec{y} = (r_1\vec{u} \oplus s_1\vec{v}) \oplus (r_2\vec{u} \oplus s_2\vec{v}) = (r_2\vec{u} \oplus s_2\vec{v}) \oplus (r_1\vec{u} \oplus s_1\vec{v}) = \vec{y} \oplus \vec{x}$
where we have used commutativity of addition in the vector space V

(2) Associativity of vector addition:

$$\begin{aligned}(\vec{x} \oplus \vec{y}) \oplus \vec{z} &= \Big((r_1\vec{u} \oplus s_1\vec{v}) \oplus (r_2\vec{u} \oplus s_2\vec{v})\Big) \oplus (r_3\vec{u} \oplus s_3\vec{v}) \\ &= (r_1\vec{u} \oplus s_1\vec{v}) \oplus \Big((r_2\vec{u} \oplus s_2\vec{v}) \oplus (r_3\vec{u} \oplus s_3\vec{v})\Big) \\ &= \vec{x} \oplus (\vec{y} \oplus \vec{z})\end{aligned}$$

where we have used associativity of vector addition in V.

(3) Existence of additive identity: We claim that $\vec{0} = 0\vec{u} \oplus 0\vec{v} \in \mathrm{Span}(\vec{u}, \vec{v})$ is an additive identity. Consider any $\vec{x} = r_1\vec{u} \oplus s_1\vec{v}$. Then we have

$$\vec{x} \oplus \vec{0} = (r_1\vec{u} \oplus s_1\vec{v}) \oplus \vec{0} = r_1\vec{u} \oplus s_1\vec{v} \oplus \vec{0} = r_1\vec{u} \oplus s_1\vec{v} = \vec{x}$$

because $\vec{0}$ is additive identity for $\oplus$ in V.

(4) Existence of additive inverse: Consider any vector $\vec{x} = r_1\vec{u} \oplus s_1\vec{v}$. We claim that $(-r_1)\vec{u} \oplus (-s_1)\vec{v}$ is the additive inverse for $\vec{x}$. This is because

$$\begin{aligned}
\vec{x} \oplus \Big((-r_1)\vec{u} \oplus (-s_1)\vec{v}\Big) &= (r_1\vec{u} \oplus s_1\vec{v}) \oplus \Big((-r_1)\vec{u} \oplus (-s_1)\vec{v}\Big) \\
&= \Big(r_1\vec{u} \oplus s_1\vec{v}\Big) \oplus (-r_1)\vec{u} \oplus (-s_1)\vec{v} \\
&= r_1\vec{u} \oplus s_1\vec{v} \oplus (-r_1)\vec{u} \oplus (-s_1)\vec{v} \\
&= r_1\vec{u} \oplus (-r_1)\vec{u} \oplus s_1\vec{v} \oplus (-s_1)\vec{v} \\
&= \Big(r_1 + (-r_1)\Big)\vec{u} \oplus \Big(s_1 + (-s_1)\Big)\vec{v} \\
&\qquad \text{(by distributivity of scalar sums in } V\text{)} \\
&= 0\vec{u} \oplus 0\vec{v} \qquad \text{(by existence of additive inverse in } F\text{)} \\
&= \vec{0} \oplus \vec{0} \quad \text{(0 times any vector is zero vector from Week 8 ex sheet)} \\
&= \vec{0} \qquad \text{(by existence of additive inverse in } V\text{)}
\end{aligned}$$

(5) Associativity of multiplication of scalar & vector:
Let $\vec{x} = r_1\vec{u} \oplus s_1\vec{v}$ and $t, q \in F$ be any two scalars. Then we have

$$\begin{aligned}
t(q\vec{x}) &= t\Big(q(r_1\vec{u} \oplus s_1\vec{v})\Big) \\
&= t\Big(qr_1\vec{u} \oplus qs_1\vec{v}\Big) && \text{(by distributivity of vector sums in } V\text{)} \\
&= (tqr_1)\vec{u} \oplus (tqs_1)\vec{v} && \text{(by associativity of multiplication of scalar \& vector in } V\text{)} \\
&= tq(r_1\vec{u}) \oplus tq(s_1\vec{v}) && \text{(by associativity of multiplication of scalar \& vector in } V\text{)} \\
&= tq(r_1\vec{u} \oplus s_1\vec{v}) && \text{(by distributivity of vector sums in } V\text{)} \\
&= tq\vec{x}
\end{aligned}$$

(6) Distributivity of scalar sums:
Let $\vec{x} = r_1\vec{u} \oplus s_1\vec{v}$ and $t, q \in F$ be any two scalars. Then we have

$$(t+q)\vec{x} = t\vec{x} \oplus q\vec{x} \qquad \text{(by distributivity of scalar sums in } V\text{)}$$

where we have used the fact that $\vec{x} = r_1\vec{u} \oplus s_1\vec{v}$ is a vector in V

(7) Distributivity of vector sums:
Let $\vec{x} = r_1\vec{u} \oplus s_1\vec{v}$ and $\vec{y} = r_2\vec{u} \oplus s_2\vec{v}$. Let $t \in \mathbb{F}$ be any scalar. Then we have

$$\begin{aligned}
t(\vec{x} \oplus \vec{y}) &= t\Big((r_1\vec{u} \oplus s_1\vec{v}) \oplus (r_2\vec{u} \oplus s_2\vec{v})\Big) \\
&= t(r_1\vec{u} \oplus s_1\vec{v} \oplus r_2\vec{u} \oplus s_2\vec{v}) \\
&= t(r_1\vec{u}) \oplus t(s_1\vec{v}) \oplus t(r_2\vec{u}) \oplus t(s_2\vec{v}) && \text{(by distributivity of vector sums in } V\text{)} \\
&= \Big(t(r_1\vec{u}) \oplus t(s_1\vec{v})\Big) \oplus \Big(t(r_2\vec{u}) \oplus t(s_2\vec{v})\Big) \\
&= \Big(t(r_1\vec{u} \oplus s_1\vec{v})\Big) \oplus \Big(t(r_2\vec{u} \oplus s_2\vec{v})\Big) && \text{(by distributivity of vector sums in } V\text{)} \\
&= t\vec{x} \oplus t\vec{y}
\end{aligned}$$

(8) Existence of identity of multiplication of scalar & vector: Since F is a field, there is a multiplicative identity 1 for multiplication of two scalars from F. We claim this 1 is also the identity for multiplication of a scalar and a vector for the vector space $\text{Span}(\vec{u}, \vec{v})$ over the field F. Let $\vec{x} = r_1\vec{u} \oplus s_1\vec{v}$ be any vector from $\text{Span}(\vec{u}, \vec{v})$. Then we have

$$\begin{aligned}
1\vec{x} &= 1(r_1\vec{u} \oplus s_1\vec{v}) \\
&= 1(r_1\vec{u}) \oplus 1(s_1\vec{v}) && \text{(by distributivity of vector sums in } V) \\
&= (1r_1)\vec{u} \oplus (1s_1)\vec{v} \\
& \text{(by associativity of multiplication of a scalar from } F \text{ and a vector from } V) \\
&= r_1\vec{u} \oplus s_1\vec{v} && \text{since 1 is a multiplicative identity for } F \\
&= \vec{x}
\end{aligned}$$

---

(4) Consider the vector space $\mathbb{Q}^3$ of 3-tuples of rational numbers over the field Q of rational numbers. Let $\vec{u} = \begin{pmatrix} 9 \\ 17 \\ 135 \end{pmatrix}, \vec{v} = \begin{pmatrix} 19 \\ 27 \\ 100 \end{pmatrix}$ and $\vec{w} = \begin{pmatrix} 28 \\ 44 \\ 235 \end{pmatrix}$. Are the vectors $\vec{u}, \vec{v}, \vec{w}$ linearly independent?

Comment: Does the observation $\vec{u} \oplus \vec{v} = \vec{w}$ help you?

Solution: Let V be a vector space over the field F. Recall that a set of vectors $\vec{v_1}, \vec{v_2}, \ldots, v_n \in V$ is linearly independent if for any scalars $r_1, r_2, \ldots, r_n \in F$

$$r_1\vec{v_1} \oplus r_2\vec{v_2} \oplus r_3\vec{v_3} \oplus \oplus .... \oplus r_n\vec{v_n} = \vec{0} \quad \Rightarrow \quad r_1 = r_2 = r_3 = ... = r_n = 0$$

If you observe that $\vec{u} \oplus \vec{v} = \vec{w}$, then this means $1\vec{u} \oplus 1\vec{v} \oplus (-1)\vec{w} = \vec{0}$. Hence, the three vectors $\vec{u}, \vec{v}$ and $\vec{w}$ are not linearly independent in the vector space $\mathbb{Q}^3$ of 3-tuples of rational numbers over the field Q of rational numbers.

If you did not observe that $\vec{u} \oplus \vec{v} = \vec{w}$, then you could find it easily through Gaussian elimination. Suppose there are three scalars $r, s, t \in \mathbb{Q}$ such that

$$r\vec{u} \oplus s\vec{v} \oplus t\vec{w} = \vec{0}$$

Then $\vec{u}, \vec{v}$ and $\vec{w}$ are linearly independent if the only solution to above equality is $r = s = t = 0$ (note that $r = s = t = 0$ is always a solution for this equality). We write the above equality explicitly as follows:

$$r\vec{u} \oplus s\vec{v} \oplus t\vec{w} = ra \begin{pmatrix} 9 \\ 17 \\ 135 \end{pmatrix} \oplus s \begin{pmatrix} 19 \\ 27 \\ 100 \end{pmatrix} \oplus t \begin{pmatrix} 28 \\ 44 \\ 235 \end{pmatrix} = \vec{0} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

Equating the first co-ordinate we get

$$9r + 19s + 28t = 0 \tag{7}$$

Equating the second co-ordinate we get

$$17r + 27s + 44t = 0 \tag{8}$$

Equating the third co-ordinate we get

$$135r + 100s + 235t = 0 \tag{9}$$

We now solve these three equations in three variables r, s, t using Gaussian elimination. Let us eliminate the variable s first.
19 multiplied by equation (9) gives

$$2565r + 1900s + 4465t = 0 \tag{10}$$

100 multiplied by equation (7) gives

$$900r + 1900s + 2800t = 0 \tag{11}$$

Equation (10) minus equation (11) gives $1665r + 1665t = 0$ which implies $t = (-r)$. Substituting this in equation (7) gives $= 9r + 19s - 28r = 19s - 19r$ which implies $r = s$. Hence, any value of r, s, t which satisfy $r = s$ and $t = (-r)$ satisfy the equation $r\vec{u} \oplus s\vec{v} \oplus t\vec{w} = \vec{0}$. In particular, we can choose $r = 1 = s$ and $t = (-1)$.